

GaussianShopVR: Facilitating Immersive 3D Authoring Using Gaussian Splatting in VR

Yulin Shen*

Boyuu Li

Jiayang Huang

Zeyu Wang

Computational Media and Arts
The Hong Kong University of Science and Technology (Guangzhou)

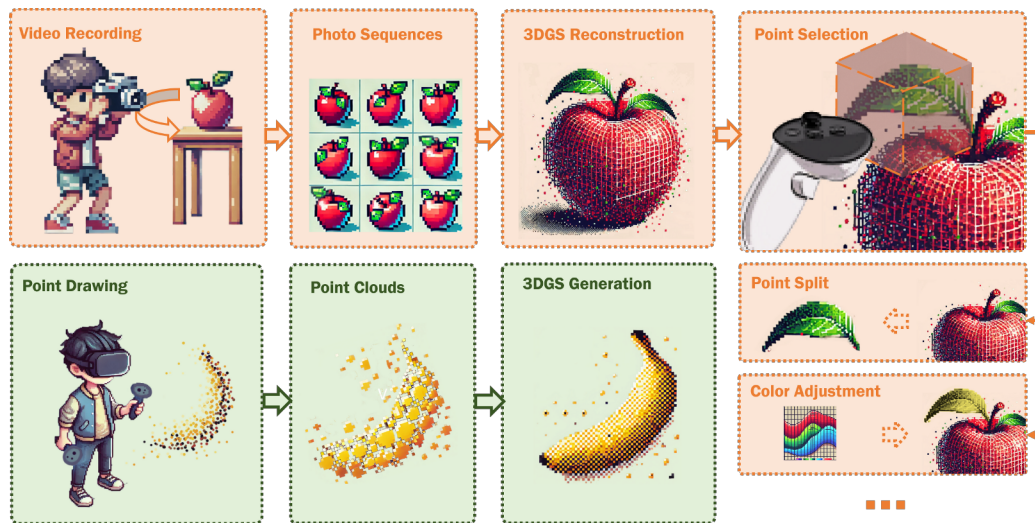


Figure 1: Workflow illustration of GaussianShopVR. Users can obtain 3D Gaussian Splatting (3DGS) objects in two ways: either by reconstructing point clouds from photo sequences or by generating them through user-drawn point clouds. Users can specify target editing areas via point selection, allowing for various editing methods to be applied to these areas.

ABSTRACT

Virtual reality (VR) applications require massive high-quality 3D assets to create immersive environments. Generating mesh-based 3D assets typically involves a significant amount of manpower and effort, which makes VR applications less accessible. 3D Gaussian Splatting (3DGS) has attracted much attention for its ability to quickly create digital replicas of real-life scenes and its compatibility with traditional rendering pipelines. However, it remains a challenge to edit 3DGS in a flexible and controllable manner. We propose GaussianShopVR, a system that leverages VR user interfaces to specify target areas to achieve flexible and controllable editing of reconstructed 3DGS. In addition, selected areas can provide 3D information to generative AI models to facilitate the editing. GaussianShopVR integrates object hierarchy management while keeping the backpropagated gradient flow to allow local editing with context information.

Index Terms: 3D authoring, VR user interface, Gaussian Splatting editing

1 INTRODUCTION

3D digital content has been ubiquitous in spatial design, animation, games, and virtual reality (VR) applications. However, 3D asset

creation remains time-consuming and labor-intensive, often requiring geometry modeling, sculpting, UV mapping, and various post-processing steps. The calls for more intuitive 3D authoring tools that could significantly improve the efficiency of asset creation. While advances in AI technologies have made 2D image generation and editing much more accessible, how to use AI to support 3D authoring remains an under-explored open research question.

We propose GaussianShopVR, a system that leverages a VR user interface for 3DGS editing. We hypothesize that VR interfaces can significantly help identify editing areas by selecting 3DGS points and user-drawn points. To control AI-driven editing, we devise a shape-control loss to leverage the spatial information of identified editing areas which constrains optimized points to be close to the user-specified shape. The identified 3D areas can also generate 3D consistent masks as input to generative AI models. To support general editing tasks, we implemented a hierarchical object structure in GaussianShopVR tailored for 3DGS, while keeping the differentiability of the whole system for AI optimization.

2 THE GAUSSIANSHOPVR SYSTEM

2.1 3D Interface

GaussianShopVR provides an immersive VR platform for 3DGS editing using a 3D user interface. This interface facilitates an intuitive and interactive experience for viewing, manipulating, and modifying 3D content. Using VR controllers, users can select and interact with not only 3DGS objects but also 3DGS points, which is a challenging task in conventional 2D interfaces [3]. With GaussianShopVR, users can also draw point clouds in VR for AI editing.

*e-mail: yshen654@connect.hkust-gz.edu.cn

2.2 Controllable AI Editing

Existing works [1, 2] focus on object-level editing without fine-grained control. With GaussianShopVR, users interact with the VR interface to select or draw points that define the areas to be edited. The corresponding 3D information is then sent to the back-end server for processing. The selected or drawn point clouds serve as constraints that guide AI models during the editing process.

2.3 Object Hierarchy

We design an object hierarchy to manage multiple objects and the whole scene consists of objects with multi-level semantics. Users can select a container object as the context for AI-driven editing, and a target object for actual editing. Though the system is complex, it keeps the gradient flow to ensure that the gradients can be back-propagated to target objects.

3 EXAMPLE TASKS

After identifying the target area by selecting Gaussian splatting points or drawing point clouds on the object, users can proceed with further edits. GaussianShopVR offers a wide range of editing features to support creative modifications.

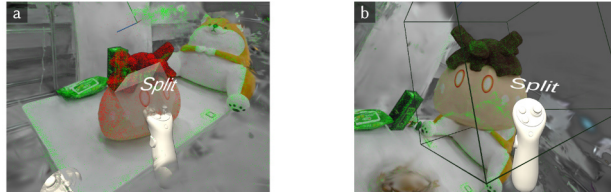


Figure 2: Splitting example.

Splitting. While 3D Gaussian Splatting effectively reconstructs entire scenes from image collections, these reconstructions are typically unified and inseparable. As shown in Fig. 2, GaussianShopVR addresses this by enabling users to split individual objects from the reconstructed scene, allowing them to extract reusable 3D assets for further editing.

Merging. In many scenarios, users may combine multiple objects to simplify scene management or create more complex models. GaussianShopVR offers a merging feature that allows users to select two objects and merge them into a single entity.

Removing. GaussianShopVR also allows users to remove unwanted sections from a 3D Gaussian Splatting scene. Similar to the splitting process, users can select specific points using the point clouds selection tool to isolate areas they wish to discard and then confirm their removal, permanently deleting the selected points from the scene. Additionally, users can directly remove the entire object from the project.



Figure 3: Inpainting example.

Inpainting. Gaps in 3D Gaussian Splatting can arise from incomplete scans or the removal and splitting of objects, leaving voids in the reconstructed scene. To address this, we support the inpainting feature. Fig. 3 shows an example where the user inpaints the void left when an object is removed from the table.

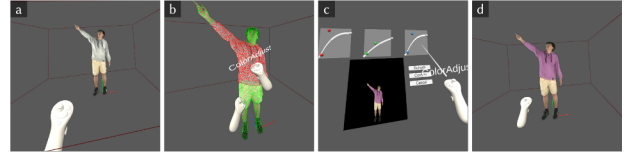


Figure 4: Color adjusting example.

Color Adjusting. The color adjustment process begins with the user selecting specific Gaussian splatting points. Once the selection is confirmed, users can proceed to adjust the RGB curves individually. A preview image is generated in real time, allowing users to see how the color changes will affect the model. After finalizing the adjustments, the color modifications are applied and rendered onto the 3D model, ensuring a seamless and accurate reflection of the selected adjustments. Fig. 4 demonstrates an example of adjusting the clothes color of a human avatar.

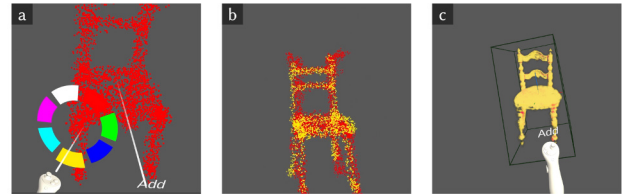


Figure 5: Generation example.

Generation. GaussianShopVR facilitates the generation of 3D Gaussian Splatting by allowing users to draw point clouds and input text prompts through voice commands. In contrast to purely text-based generation, the incorporation of 3D point clouds provides spatial geometry guidance, enabling users to exert greater control over the structure of the generated content. As shown in Fig. 5, the user can draw a colored point cloud and provide a text prompt via voice input as a condition for 3DGS generation.

4 CONCLUSION

GaussianShopVR aims to enhance immersive 3D authoring by adopting the 3D Gaussian Splatting representation, leveraging its advantages in delivering immersive environments through photorealistic, high-FPS rendering, and seamless integration with AI editing. GaussianShopVR offers an intuitive and accurate selection of point clouds and provides 3D information for AI-driven editing control, while existing methods relying on 2D user interfaces often fall short in specifying editing areas and controlling outcomes.

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